

# Tz-Ying (Gina) Wu

## Work Experience

- Present **Staff AI Research Scientist, Intel Multimodal AI**
- Leading applied research projects on multimodal and agentic AI systems, focusing on the design of scalable architectures and end-to-end workflows for enterprise applications.
- Aug. 2024 - **AI Research Scientist, Intel AI Lab**
- Apr. 2026
- Conducted research on multimodal understanding, building and fine-tuning VLMs and multimodal LLMs for advanced cross-modal reasoning, leading to multiple publications and technology transfer.
  - Developed inference-time scaling techniques for video generation with enhanced motion control.
  - Collaborated cross-functionally to deploy research innovations into production systems, including multimodal pipelines for video search and summarization on Intel's AI platform. [\[Link\]](#)
- Jun. 2023 - **Machine Learning Research Intern, Intel AI Lab**
- Mar. 2024
- Proposed a lightweight adaptation method for video foundation models, achieving high fine-tuning performance by optimizing less than 1% of model parameters. Research published at WACV 2025.
- Jun. 2021 - **Applied Scientist Intern, Amazon AWS AI Lab**
- Sep. 2021
- Developed an efficient solution for large-scale class-incremental learning by addressing the stability-plasticity dilemma and mitigating data imbalance. Research published at CVPR 2022.
- Jul. 2015 - **Signal Processing Intern, IMEC-Taiwan CO. R&D department**
- Aug. 2015
- Designed and implemented neuron-signal generator, encoder, and decoder framework (Python/C++), improving the real-time decoding speed by at least 2 times.

## Education

- Sep. 2018 - **University of California San Diego (UCSD), La Jolla, CA**
- Jun. 2024
- Ph.D. in Electrical Computer Engineering
- Advisor: Professor Nuno Vasconcelos
- Feb. 2016 - **National Tsing Hua University (NTHU), Hsinchu, Taiwan**
- Jan. 2018
- M.S. in Electrical Engineering, GPA: 4.27/4.3
- Advisor: Professor Min Sun
- Sep. 2012 - **National Tsing Hua University (NTHU), Hsinchu, Taiwan**
- Jan. 2016
- B.S. in Electrical Engineering, GPA: 4.02/4.3, Ranking: 10/99 (graduated a half-year earlier)

## Research Interest

**Deep Learning & Computer Vision:** Publication in top-tier conferences across broad topics, including visual-language model, video/scene understanding, model robustness, and efficient transfer.

## Publications

- 2026 Sainan Liu\*, **Tz-Ying Wu\***, Hector A Valdez\*, Subarna Tripathi, "Search2Motion: Training-Free Object-Level Motion Control via Attention-Consensus Search", *Preprint*, 2026 [\[Paper\]](#)
- 2026 Shubhashis Roy Dipta\*, **Tz-Ying Wu\***, Subarna Tripathi, "VC-Inspector: Advancing Reference-free Evaluation of Video Captions with Factual Analysis", In *The 64th Annual Meeting of the Association for Computational Linguistics (ACL main)*, 2026 [\[Paper\]](#)
- 2026 **Tz-Ying Wu**, Sharath Nittur Sridhar, Subarna Tripathi, "Harnessing Object Grounding for Time-Sensitive Video Understanding", In *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2026 [\[Paper\]](#)

- 2025 **Tz-Ying Wu\***, Tahani Trigui\*, Sharath Nittur Sridhar, Anand Bodas, Subarna Tripathi, "Toward Scalable Video Narration: A Training-free Approach Using Multimodal Large Language Models", In *IEEE International Conference on Computer Vision (ICCV) Workshop*, 2025 [Paper]
- 2025 Ivan Rodin\*, **Tz-Ying Wu\***, Kyle Min, Sharath Nittur Sridhar, Antonino Furnari, Subarna Tripathi, Giovanni Maria Farinella, "EASG-Bench: Video Q&A Benchmark with Egocentric Action Scene Graphs", In *IEEE International Conference on Computer Vision (ICCV) Workshop*, 2025 [Paper]
- 2025 **Tz-Ying Wu**, Kyle Min, Subarna Tripathi, Nuno Vasconcelos, "Ego-VPA: Egocentric Video Understanding with Parameter-efficient Adaptation", In *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2025 [Paper]
- 2024 **Tz-Ying Wu\***, Chih-Hui Ho\*, Nuno Vasconcelos, "ProTeCt: Prompt Tuning for Taxonomic Open Set Classification", In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024 [Paper]
- 2022 Alakh Desai, **Tz-Ying Wu**, Subarna Tripathi, Nuno Vasconcelos, "Single-Stage Visual Relationship Learning using Conditional Queries", In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022 [Paper]
- 2022 **Tz-Ying Wu**, Gurumurthy Swaminathan, Zhizhong Li, Avinash Ravichandran, Nuno Vasconcelos, Rahul Bhotika, Stefano Soatto, "Class-Incremental Learning with Strong Pre-trained Models", In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022 [Paper]
- 2021 Alakh Desai\*, **Tz-Ying Wu\***, Subarna Tripathi, Nuno Vasconcelos, "Learning of Visual Relations: The Devil is in the Tails", In *IEEE International Conference on Computer Vision (ICCV)*, 2021 [Paper]
- 2020 **Tz-Ying Wu**, Pedro Morgado, Pei Wang, Chih-Hui Ho, Nuno Vasconcelos, "Solving Long-tailed Recognition with Deep Realistic Taxonomic Classifier", In *European Conference on Computer Vision (ECCV)*, 2020 [Paper]
- 2020 Yiran Xu, Xiaoyin Yang, Lihang Gong, Hsuan-Chu Lin, **Tz-Ying Wu**, Yungsheng Li, Nuno Vasconcelos, "Explainable Object-induced Action Decision for Autonomous Vehicles", In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020 [Paper]
- 2020 Chih-Hui Ho, Bo Liu, **Tz-Ying Wu**, Nuno Vasconcelos, "Exploit Clues from Views: Self-Supervised and Regularized Learning for Multiview Object Recognition", In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020 [Paper]
- 2018 **Tz-Ying Wu\***, Juan-Ting Lin\*, Tsun-Hsuang Wang, Chan-Wei Hu, Juan Carlos Niebles, Min Sun, "Liquid Pouring Monitoring via Rich Sensory Inputs", In *European Conference on Computer Vision (ECCV)*, 2018 [Paper]
- 2017 **Tz-Ying Wu\***, Ting-An Chien\*, Cheng-Sheng Chan, Chan-Wei Hu, Min Sun, "Anticipating Daily Intention using On-Wrist Motion Triggered Sensing", In *IEEE International Conference on Computer Vision (ICCV)*, **Spotlight**, 2017 [Paper]

## --- Patents

- 2025 US Patent No. 12406469 – "Class-incremental learning with pretrained machine learning models", *Granted*, 2025 [Link]

## --- Demos & Presentations

- 2025 Sainan Liu\*, Hector A Valdez\*, **Tz-Ying Wu\***, Sameer Sheorey, Kyle Min, Diana Wofk, Benjamin Ummenhofer, Michael Paulitsch, Subarna Tripathi, "Edit2Motion: Towards Training-Free Interactive Object-Level Motion Guidance for Image-to-Video Generation", In *IEEE International Conference on Computer Vision (ICCV) Workshop*, 2025 [Demo]
- 2016 Cheng-Sheng Chan, Ting-An Chien, **Tz-Ying Wu**, Min Sun, "Recognition from Hand Cameras: A Revisit with Deep Learning", In *Asian Conference on Computer Vision*, 2016 [Demo1, Demo2]

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## Invited Talks

- 2026 "Interpretability as Factual Diagnosis for Video Captioning – Reference-Free Evaluation with Explanations", In EnCORE Workshop on Interpretability in Modern AI - Foundations, Methods, and Emerging Directions. [Link]

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## Academic Services

- Reviewer
- **Journal:** TPAMI, IJCV, TNNLS, Neurocomputing.
  - **Conference:** CVPR (*outstanding reviewer in 2022, 2026*), ICCV, ECCV, NeurIPS (*top reviewer in 2025*), ACL (*good review*), ICLR, ICML, WACV, AAAI, ICASSP, ACM-MMM.
  - **Workshop:** ECCV Imbalance Problems in Computer Vision Workshop, NeurIPS Women in Machine Learning Workshop, ICLR Reliable and Responsible Foundation Models Workshop, AAAI AI with Biased or Scarce Data Workshop, ECCV Women in Computer Vision Workshop, AAAI Scalable and Efficient Artificial Intelligence Systems Workshop, ICLR Scalable Optimization for Efficient and Adaptive Foundation Models Workshop, NeurIPS Lock-LLM Workshop
- Organizer ICCV 2025 Computer Vision in Advertising and Marketing Workshop
- Program ECCV 2020 Imbalance Problems in Computer Vision Workshop, AAAI 2024 AI with Biased or Scarce Data Workshop
- Committee
- Volunteer CVPR 2020 AC meeting, ICCV 2017, ACCV 2016.

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## Honors and Distinctions

- 2026 CVPR 2026 Outstanding Reviewer Award
- 2025 NeurIPS 2025 Top Reviewer
- 2024 CVPR 2024 Doctoral Consortium
- 2022 Amazon Post-Internship Fellowship
- 2022 CVPR 2022 Outstanding Reviewer Award
- 2018 UCSD Electrical and Computer Engineering Department Fellowship
- 2017 Appier Scholarship for Attending International Conference
- 2017 NTHU EECS Scholarship for Attending International Conference
- 2017 CVGIP Excellent Paper Award
- 2016 NTHU EE Admission Scholarship
- 2014 NTHU EECS Scholarship for Outstanding Students
- 2014 Jia-Bin Chen Scholarship for Outstanding Students in EE
- 2013 NTHU Scholarship for Academic Excellence
- 2012 - 2013 Breaking records in athletics of NTHU Track&Field on three occasions

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## Leadership

- 2019 - 2023 **Mentor of Summer Research Internship Program (SRIP), UCSD**
- Mentored BS/MS students on deep learning and computer vision research.
  - **2019 - 2020** Collected the *BDD-OIA* dataset, a large-scale object-induced action dataset for autonomous driving, as an important add-on to the BDD dataset. Published a paper at the CVPR 2020 conference.
  - **2021 - 2022** Researched on scene graphs generation. Published two papers at the ICCV 2021 and NeurIPS 2022, respectively.
  - **2022 - 2023** Organized a study group for BEV (Bird's-Eye-View) object detection.
- 2022 - 2023 **Mentor of UCSD Graduate Women in Computing (GradWic), UCSD**
- Held regular meetings with mentees to discuss questions in graduate studies.
- 2012 - 2013, **Member of EE Student Association of Academic Affair, NTHU**
- 2014 - 2015
- Organized outreach events for high-school students and talks for the department.

- 2014 - 2015 **Team Leader in the NTHU EE Women's Basketball Team, NTHU**
- Held recruiting and team-building events, and organized inter-school games registration.
- 2014 **Member of Tsinghua Friendship Ambassadors Association (TFAA), NTHU**
- Assisted in holding the 2014 NTHU International Week.

## Research Experience

- Jan. 2018 - **Graduate Student Researcher, Statistical Visual Computing Lab, UCSD**
- Jun. 2024
- Worked on long-tailed recognition, continual learning, and scene understanding problems.
  - Focused on generating reliable predictions and adapting to continually changing distributions.
  - Published 7 papers in top-tier venues, including the release of the *BDD-OIA* dataset.
- Feb. 2016 - **Graduate Research Assistant, Vision Science Lab, NTHU**
- Jan. 2018
- Developed wearable camera and IOT system with Raspberry Pi embedded boards.
  - Conducted deep learning experiments for learning human behaviors using multi-modal fusion.
  - Published 2 papers in top-tier venues, including the release of the *Daily Intention* dataset.
- Sep. 2014 - **Undergraduate Research, advised by Prof. Jing-Jia Liou, NTHU**
- Jun. 2015
- Conducted research on FPGA Design of Accelerators with High-level Evaluation and Synthesis Tools.
  - Accelerated the runtime of the opensource software *ALADDIN*. [Github PR]

## Teaching Experience

- UCSD
- ECE 228 Machine Learning for Physical Applications, *Spring 2023 & 2024*
  - MAE 146 Introduction to Machine Learning Algorithms, *Winter 2023*
  - ECE 271A Statistical Learning I, *Fall 2021 & 2022*
  - ECE 175A Elements of Machine Intelligence, *Winter 2020*
  - ECE 271B Statistical Learning II, *Winter 2019*
- NTHU
- Deep Learning Lab, *UNITEC - NTHU Summer School 2017*
  - EECS2020 Signals and Systems, *Spring 2016 & 2017*
  - EE2310 Introduction to Programming (C/C++), *Fall 2015*

## Skills

- Programming Python, C/C++, MATLAB, Verilog, Assembly, HTML,  $\LaTeX$
- Library PyTorch, HuggingFace, vLLM, SGLang, Gradio, OpenCV, Scikit-learn, Pandas, Matplotlib
- System & Tool Linux, Git, Kubernetes, AWS, Azure, Docker
- Language Mandarin, English